

DAXHIEL PERUGORRIA RUCIEL

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EDUCATION

University of Texas at Austin – Austin, TX

2025 – 2029

Bachelor of Science in Computer Science

Coursework: Data Structures and Algorithms, Computer Systems, Computer Organization and Architecture, Discrete Mathematics, Linear Algebra, Probability and Statistics

PROJECTS

qumidi | *Open-Source Python Library*

- **Reverse-engineered and implemented** the Allen & Heath Qu-series MIDI-over-TCP protocol into a zero-dependency Python library, exposing parity sync/async APIs (QuMixer/AsyncQuMixer) with a tiered parameter registry that flags hardware-unverified values, avoiding returning plausible-looking defaults.
- **Engineered a hardware-free test harness**, pairing a fake console emulator with golden-vector fixtures to round-trip encode/decode every protocol parameter, driving 86 tests to a 90% coverage gate on core protocol/state modules and powering full CLI tooling (probe, monitor, record/replay) without physical mixer access.

Recommendation Engine Evaluation Model | *Datahack 2026 Finalist*

- **Architected a causal Transformer skip-prediction model** for sequential listening behavior by converting each session into a token stream (user attributes + track semantics + context + labels), enabling next-step prediction of the final skip label and reaching 87.07% validation skip accuracy during training.
- **Scaled representation learning to over 1 million tracks** by generating 3-level hierarchical “semantic” track IDs via MiniBatchKMeans clustering ($32 \times 32 \times 32$) on standardized audio features, then encoding each track with clustered IDs + binned popularity/skip-rate signals to improve generalization on long-tail content.

VibeDungeon | *AI Agent-Driven Game*

- **Designed and built a modular Python RPG engine** with AI-generated encounters and procedurally-generated dungeon maps, along with isolated combat, character, and dungeon systems behind clean interfaces.
- **Integrated an LLM-driven agent layer** by introducing provider-agnostic agent abstractions, resulting in AI-generated narrative content isolated from core game logic.

rustml | *ML Systems Library, Rust*

- **Architected an autograd engine and neural network library in Rust**, implementing reverse-mode automatic differentiation over a dynamically-built computation graph with Rc<RefCell<Tensor>>-based shared ownership, deriving forward/backward rules for 15+ tensor ops (matmul, softmax, layer norm, embedding lookup) to support DAG-structured graphs with shared subgraphs.
- **Implemented a full transformer architecture from primitive tensor operations** (multi-head attention, layer normalization, and feed-forward blocks) alongside SGD/Adam optimizers and a PyTorch-style DataLoader, validating correctness against a PyTorch baseline on MNIST and covering the engine with 84 unit tests spanning autograd, layers, and optimizers.

EXPERIENCE

Project Advance Austin – Austin, TX

August 2026 – Present

Data Analytics Consultant

- **Engineering an interactive geospatial analytics tool** in Tableau, mapping geographic coordinates and relational distances between incarcerated mothers and their children across Texas and the U.S. to drive nonprofit fundraising.
- **Ingested, cleaned, and structured relational database records** of families served, resolving anomalies in geographic, demographic, and facility data.
- **Synthesizing multi-source empirical literature and program metrics** into an online data-driven publication, providing technical documentation to support data-driven grant and corporate sponsorship acquisition.

Code2College – Remote

June 2026 – July 2026

Course Instructor

- **Instructed 24+ students** over an 18-session curriculum, achieving 95% average attendance and 97% average quiz scores.
- **Taught SDE fundamentals**, including Python programming, object-oriented design, error handling, and basic data management.

Getfertilizerwiser – Austin, TX

February 2026 – May 2026

Freelance Web Developer

- **Architected a full-stack Next.js/React site** for a nonprofit client, engineering a token-driven Tailwind design system and a reusable component library (layout, section, and UI primitives) to keep 7 client-facing pages visually consistent without a CMS.
- **Structured client-provided content into typed static data models** (team bios, events, sponsors, resources) and integrated Formspre and Cloudinary for serverless form handling and media delivery, shipping the site to production on Vercel.

TECHNICAL SKILLS

- **Tools & Skills:** Pytorch, FastAPI, Next.js, React, Vercel, Supabase, Express, SQL, Data Visualization, Geospatial Analysis,
- **Languages:** Python, Java, Rust, Javascript, C/C++
- **Work Eligibility:** Eligible to work in the U.S. with no restrictions